

Chapter 6 Notes: Inverse Laplace Transforms

1. Why Go Back to the Time Domain?

So far, the course has gone **time domain** → **Laplace domain** so we could:

1. Build block diagrams
2. Do basic analysis (Initial Value Theorem, Final Value Theorem, finding zeros/poles)
3. Work with open-loop and closed-loop models

This chapter reverses that process: **Laplace domain** → **time domain**, so we can get an actual $y(t)$ that describes how the system behaves over time. The tool for this is a **Look-Up Table (LUT)** combined with **partial-fraction expansion (PFE)**.

2. Requirement: Strictly Proper Transfer Functions

PFE requires the transfer function to be **strictly proper**: the numerator's order must be *less than* the denominator's order.

- **If it's not strictly proper** (numerator order $\geq$ denominator order), do **long division once**. This splits the transfer function into:

a constant + a strictly proper remainder

Then apply PFE only to the strictly proper remainder.

Convention: write the denominator in **binomial form with real roots** (i.e., factored into $(s + p_1)(s + p_2) \cdots$). Any **complex roots** are left as an unfactored **second-order expression** (since complex roots come in conjugate pairs and factoring them individually would reintroduce complex numbers).

3. The Three PFE Forms

(a) Distinct real poles only

$$\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)} = \frac{k_1}{s + p_1} + \frac{k_2}{s + p_2} + \cdots + \frac{k_n}{s + p_n}$$

(b) Real poles + one pair of complex poles

The complex pair stays together as a 2nd-order denominator term, with a **linear numerator** ($k_m s + k_{m+1}$) instead of a constant:

$$\frac{Y(s)}{R(s)} = \frac{k_1}{s + p_1} + \frac{k_2}{s + p_2} + \cdots + \frac{k_m s + k_{m+1}}{\alpha_2 s^2 + \alpha_1 s + \alpha_0}$$

(c) Repeated poles (real and/or complex)

Each repeated pole needs **one term per multiplicity**, with increasing denominator powers:

$$\frac{Y(s)}{R(s)} = \frac{k_1}{(s + p_1)^2} + \frac{k_2}{s + p_1} + \frac{k_3}{s + p_2} + \cdots + \frac{k_m s + k_{m+1}}{(\alpha_2 s^2 + \alpha_1 s + \alpha_0)^2} + \frac{k_{m+2} s + k_{m+3}}{\alpha_2 s^2 + \alpha_1 s + \alpha_0}$$

Pattern to notice: if a real pole p_1 is repeated twice, you get *two* terms for it — one with $(s + p_1)^2$ in the denominator and one with just $(s + p_1)$. Same logic extends to complex repeated poles, but with the quadratic denominator squared plus a single-power version.

Caveat: PFE gets harder — and less numerically reliable — as systems get more complex. Many repeated poles, or poles that are very close together, produce **ill-conditioned** systems (small numerical errors get amplified). Most real physical systems, though, can be solved accurately with PFE.

4. Worked Example 1 — Distinct Real Poles, Impulse Input

Given:

$$\frac{Y(s)}{R(s)} = \frac{3}{(s + 6)(s + 2)}, \quad r(t) = \delta(t)$$

Since $R(s) = 1$ for an impulse, $Y(s)$ is just the transfer function itself. Set up the partial fraction form:

$$Y(s) = \frac{3}{(s + 6)(s + 2)} = \frac{k_1}{s + 6} + \frac{k_2}{s + 2}$$

****Solving for k_1 **** — the "cover-up" method: multiply both sides by $(s + 6)$, then plug in $s = -6$ (the root of that factor). This makes the k_2 term vanish because it still has an $(s + 6)$ factor multiplying it:

$$\left. \frac{3}{s + 2} \right|_{s=-6} = k_1 + \underbrace{\frac{k_2(s + 6)}{s + 2}}_{=0 \text{ at } s=-6}$$

$$k_1 = \frac{3}{-4} = -\frac{3}{4}$$

****Solving for k_2 **** — same trick, multiply by $(s + 2)$ and plug in $s = -2$:

$$\left. \frac{3}{s + 6} \right|_{s=-2} = \underbrace{\frac{k_1(s + 2)}{s + 6}}_{=0 \text{ at } s=-2} + k_2$$

$$k_2 = \frac{3}{4}$$

Result:

$$Y(s) = \frac{-3/4}{s + 6} + \frac{3/4}{s + 2}$$

Apply the LUT (recall $\frac{1}{s+a} \leftrightarrow e^{-at}$):

$$y(t) = -\frac{3}{4}e^{-6t} + \frac{3}{4}e^{-2t}$$

This is valid for $0 \leq t < \infty$.

The "cover-up" method, generalized: to find k_i for a distinct pole p_i , multiply both sides by $(s + p_i)$, then substitute $s = -p_i$. Every other term vanishes because it still contains the factor $(s + p_i)$ evaluated at its own root.

5. Worked Example 2 — Biproper Transfer Function, Impulse Input

Given:

$$\frac{Y(s)}{R(s)} = \frac{s^2 + 13s + 6}{s^2 + 9s + 8}, \quad r(t) = \delta(t)$$

Numerator and denominator are the **same order** (2nd), so this is **not strictly proper** — long division is required first.

Step 1 — Long division:

$$Y(s) = 1 + \frac{4s - 2}{s^2 + 9s + 8}$$

(The "1" is the quotient; $\frac{4s-2}{s^2+9s+8}$ is the strictly-proper remainder.)

Step 2 — Factor the remainder's denominator and apply PFE:

$$\frac{4s - 2}{(s + 8)(s + 1)} = \frac{k_1}{s + 8} + \frac{k_2}{s + 1}$$

Solve k_1 (cover-up, $s = -8$):

$$k_1 = \left. \frac{4s - 2}{s + 1} \right|_{s=-8} = \frac{4(-8) - 2}{-8 + 1} = \frac{-34}{-7} = \frac{34}{7}$$

Solve k_2 (cover-up, $s = -1$):

$$k_2 = \left. \frac{4s - 2}{s + 8} \right|_{s=-1} = \frac{4(-1) - 2}{-1 + 8} = \frac{-6}{7}$$

Result:

$$Y(s) = 1 + \frac{34/7}{s + 8} + \frac{-6/7}{s + 1}$$

Apply the LUT (recall constant "1" in Laplace domain $\rightarrow$ impulse $\delta(t)$ in time domain; $\frac{1}{s+a} \rightarrow e^{-at}$):

$$y(t) = \delta(t) + \frac{34}{7}e^{-8t} - \frac{6}{7}e^{-t}$$

Key lesson: the "1" left over from long division transforms back into an **impulse function**, not a plain constant — because $\mathcal{L}\{\delta(t)\} = 1$.

6. Worked Example 3 — Repeated Real Poles, Step (Setpoint) Input

Given:

$$\frac{Y(s)}{R(s)} = \frac{s + 72}{s^3 + 14s^2 + 57s + 72}, \quad r(t) = 1 \text{ (step input)}$$

Since $r(t) = 1$ is a step, $R(s) = \frac{1}{s}$, so $Y(s) = \frac{Y(s)}{R(s)} \cdot R(s)$, which introduces an **extra factor of s in the denominator**. Factoring the denominator (including that extra s) gives:

$$Y(s) = \frac{s + 72}{s(s + 3)(s + 3)(s + 8)} = \frac{k_1}{s} + \frac{k_2}{(s + 3)^2} + \frac{k_3}{s + 3} + \frac{k_4}{s + 8}$$

Notice the **repeated pole at $s = -3$** requires two terms: $\frac{k_2}{(s + 3)^2}$ and $\frac{k_3}{s + 3}$.

Residues solvable by the ordinary cover-up method: $k_1 = 1, k_2 = -4.6, k_4 = -0.32$ (each found the same way as in Example 1 — multiply by the matching factor, plug in the root).

The tricky one — k_3 (the "extra" term for a repeated pole): Cover-up alone can't isolate k_3 because it's tangled up with k_2 's $(s + 3)^2$ term. Instead:

1. Multiply both sides by $(s + 3)^2$ (clearing the *highest* power of the repeated factor):

$$k_3 = \frac{d}{ds} \left[(s + 3)^2 \cdot \frac{s + 72}{s(s + 3)^2(s + 8)} \right]_{s=-3} = \frac{d}{ds} \left[\frac{s + 72}{s(s + 8)} \right]_{s=-3}$$

2. **Take the derivative** with respect to s (using the Quotient Rule — see box below), *then* substitute $s = -3$:

$$k_3 = \left. \frac{(1)(s^2 + 8s) - (s + 72)(2s + 8)}{(s^2 + 8s)^2} \right|_{s=-3}$$

3. Plug in $s = -3$ step by step:

$$k_3 = \frac{((-3)^2 - 24) - (72 - 3)(2(-3) + 8)}{((-3)^2 - 24)^2} = \frac{(9 - 24) - (69)(2)}{(9 - 24)^2} = \frac{-15 - 138}{225} = \frac{-153}{225} = -0.68$$

Result:

$$Y(s) = \frac{1}{s} - \frac{4.6}{(s + 3)^2} - \frac{0.68}{s + 3} - \frac{0.32}{s + 8}$$

Apply the LUT (recall $\frac{1}{s} \rightarrow 1$ (step), $\frac{1}{(s+a)^2} \rightarrow te^{-at}$, $\frac{1}{s+a} \rightarrow e^{-at}$):

$$y(t) = 1 - 4.6te^{-3t} - 0.68e^{-3t} - 0.32e^{-8t}$$

Key lesson — the repeated-pole recipe: for a pole repeated r times, $(s + p)^r$, you get r separate partial-fraction terms with powers $(s + p)^r, (s + p)^{r-1}, \dots, (s + p)^1$. The residue for the **highest power** term (k_2 here) is found by ordinary cover-up. The residue for **each lower power** (k_3 here) requires taking successive derivatives of the covered-up expression before substituting the root — this is a form of the general residue formula for repeated poles.

★ **Quotient Rule (given as a reference in the chapter)**

$$f(x) = \frac{g(x)}{h(x)} \Rightarrow f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{h(x)^2}$$

7. Section 6.1 — Doing PFE in MATLAB

Manually finding residues (especially for repeated or complex poles) is tedious — MATLAB's `residue` function automates it.

Syntax:

matlab

```
[r, p, k] = residue(B, A)
```

- `B` = numerator coefficients (highest power first)
- `A` = denominator coefficients (highest power first)
- Returns: `r` = residues, `p` = poles, `k` = direct term (from long division, if any — empty `[]` if the system was strictly proper)

Applying it to Example 3:

$$\frac{Y(s)}{R(s)} = \frac{s + 72}{s^3 + 14s^2 + 57s + 72}$$

- `B = [1, 72]` (from $s + 72$)
- `A = [1, 14, 57, 72, 0]` — **note the trailing 0:** since the input is a step (setpoint), the extra factor of s from $R(s) = 1/s$ must be multiplied into A before calling `residue`.

matlab

```
>> [r,p,k] = residue(B,A)
r =
    -0.3200
    -0.6800
    -4.6000
     1.0000
p =
    -8.0000
    -3.0000
    -3.0000
     0
k =
    []
```

These residues/poles match the hand-calculated values from Example 3 exactly (in a different order): $k_4 = -0.32$ at $p = -8$, $k_3 = -0.68$ at $p = -3$, $k_2 = -4.6$ at the repeated $p = -3$, and $k_1 = 1$ at $p = 0$. The empty $\boxed{k}$ confirms the transfer function (after distributing the input) was already strictly proper, so no long-division constant term was needed.

Practical tip: always remember to fold the input signal's own poles (e.g., the s from a step, or an s^2 from a ramp) into $\boxed{A}$ before calling `residue`, since `residue` operates on $Y(s)$ as a whole, not just the open-loop transfer function.

Quick-Reference Summary

- **Goal:** convert $Y(s)$ back into $y(t)$ using PFE + LUT.
- **Must be strictly proper** first — if not, do long division once; the leftover constant becomes an **impulse** $\delta(t)$ in the time domain.
- **Three PFE templates:** distinct real poles $\rightarrow$ simple fractions; complex pole pairs $\rightarrow$ keep as 2nd-order term with linear numerator; repeated poles $\rightarrow$ one term per power, from highest power down to first power.
- **Cover-up method:** multiply by the target pole's factor, substitute its root, to get that residue — works directly for the *highest-power* term of each pole.
- **Repeated-pole residues at lower powers** require differentiating (via Quotient Rule) before substituting the root.
- **MATLAB's** `residue(B,A)` automates all of this — just remember to fold the input's own transform (e.g. the s from a step) into $\boxed{A}$ first.
- **Watch for ill-conditioning:** many repeated or closely-spaced poles can make PFE numerically unreliable, though this is rare for typical physical systems.